

$$\text{result} = \alpha W + \beta B$$

$$\text{row of } A = [W, B]$$

$$X = (A^T A)^{-1} \cdot A^T b$$

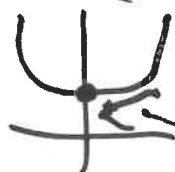
Formula for least Squares

$$A x = b$$

$$\text{Minimize } \|Ax - b\|^2$$

$$\sigma = \frac{d}{dt} \langle A(x + tu) - b, A(x + tu) - b \rangle$$

at $t=0$

 - Parabola
min at $t=0$

$$0 = \langle Au, Ax - b \rangle$$

$$= \langle u, A^T Ax - A^T b \rangle$$

true for all u

$$\Rightarrow A^T Ax - A^T b = 0$$

$$X = (A^T A)^{-1} A^T b$$

if $A^T A$ is invertible

$$A x = b$$

$$\text{e.g. } a_{1,1} x_1 + a_{1,2} x_2 + \dots + a_{1,n} x_n = b_1$$

$$\vdots$$

$$\text{e.g. } 2x_1 + 3x_2 = 4$$

$$5x_1 + 6x_2 = 7$$

$$A = \begin{pmatrix} 2 & 3 \\ 5 & 6 \end{pmatrix} \quad b = \begin{pmatrix} 4 \\ 7 \end{pmatrix}$$

Least Squares

$$\text{Want } x_1 \sin 8 + x_2 \cos 8 = 9$$

$$x_1 \sin 10 + x_2 \cos 10 = 11$$

$$x_1 \sin 12 + x_2 \cos 12 = 14$$

$$\text{Want } Ax = b \quad A = \begin{pmatrix} \sin 8 & \cos 8 \\ \sin 10 & \cos 10 \\ \sin 12 & \cos 12 \end{pmatrix} \quad b = \begin{pmatrix} 9 \\ 11 \\ 14 \end{pmatrix}$$

Use least Squares

$$y = x_1 \sin t + x_2 \cos t$$

Want Formula

$$y = \alpha \sin t + \beta \cos t + \gamma/t$$

Want α, β, γ

$$Ax = b$$

$$x = \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}$$

Data:

y	t	$\sin t$	$\cos t$	$1/t$
2.1	0.5	0.4	0.7	2
1.6	1	0.6	0.4	1
2.4	2	0.8	-0.2	0.5
2.3	3	0.1	-0.9	0.33

Want

$$2.1 = \alpha(0.4) + \beta(0.7) + \gamma(2)$$

$$Ax = b$$

Computer gives

$$x = \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}$$

$$A = \begin{pmatrix} 0.4 & 0.7 & 2 \\ 0.6 & 0.4 & 1 \\ 0.8 & -0.2 & 0.5 \\ 0.1 & -0.9 & 0.33 \end{pmatrix} \quad b = \begin{pmatrix} 2.1 \\ 1.6 \\ 2.4 \\ 2.3 \end{pmatrix}$$